

The Quantum World

KKKC6353 Nanoelectronic Devices · Lecture 2, Part 1 of 3

Dr Muhammad Faiz Bukhori

Universiti Kebangsaan Malaysia · 2026

Lecture 2 in three parts

Part 1 · The quantum world

Tiny things behave as waves. A trapped wave can only have certain energies, and the smaller the trap, the further apart they are.

Part 2 · Wells, wires and dots

Trapping electrons in one, two or three directions, and the lasers, transistors, nanotubes and screens it makes possible.

Part 3 · Making them

Carving structures top-down, growing them bottom-up, and making quantum dots and nanotubes.

Small enough to change colour

- Every vial holds the **same material**
- Only the **size** of the crystals differs: a few nanometres
- Smallest glow **violet**, largest glow **red**
- These are **quantum dots**, already in TV screens and medical imaging
- Ordinary physics cannot explain this. By the end of Part 1 we can



Photograph: Antipoff, Wikimedia Commons (CC BY-SA 3.0).

1

Two worlds: classical and quantum

Classical physics works beautifully for apples and planets, and fails for atoms.

The physics of the everyday

Classical physics pictures a predictable, clockwork universe. It rests on three pillars.

Newtonian mechanics

Motion and forces: falling apples, cars, rockets.

Electromagnetism

Electricity, magnetism and light.

Thermodynamics

Heat, energy and engines.

Where it fails

The very fast (relativity) and the very small: atoms and electrons need quantum mechanics.

Why the planetary atom fails

- Picture: electrons orbiting a nucleus like planets
- A charge moving in a circle **gives off light**
- So it should lose energy and **spiral in** within about 10^{-11} s
- But atoms are **stable**, with only certain energies
- At this scale classical physics is simply wrong

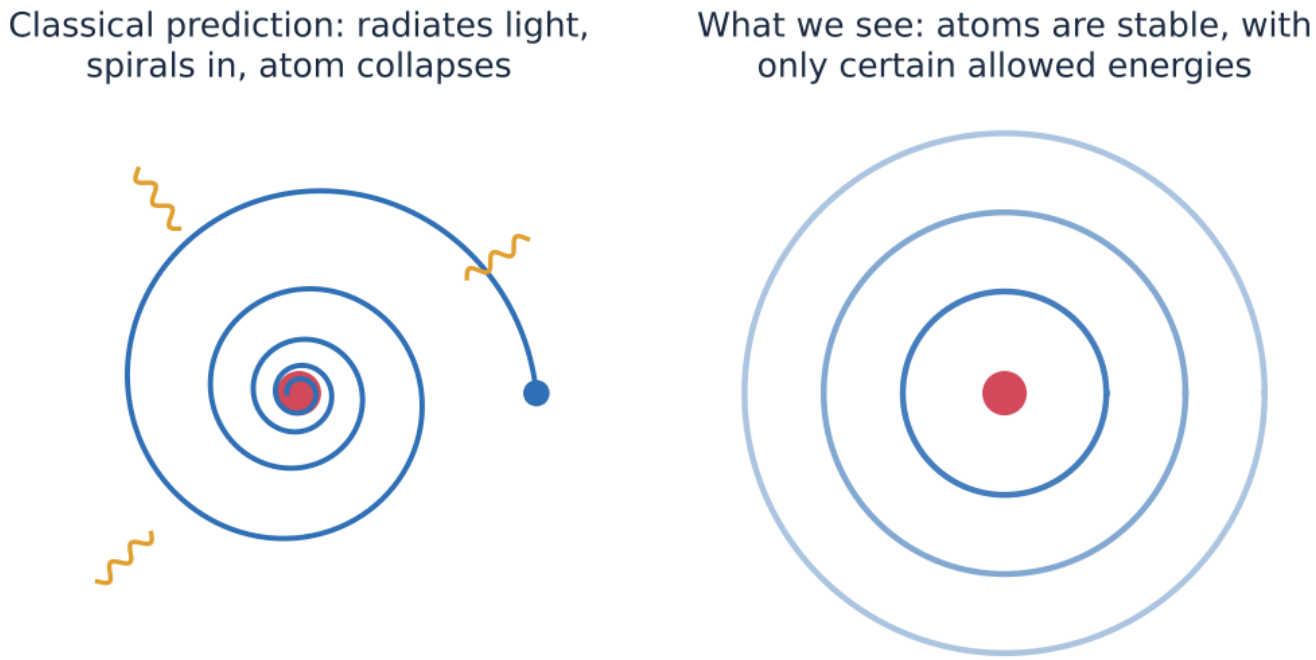


Figure made for this course.

2

Light and matter: waves and particles

Is light a wave or a stream of particles? Yes.

Light arrives as particles

- Shine light on a metal and electrons can be knocked out
- **Bright red light**: nothing comes out
- **Dim blue light**: electrons fly out at once
- Einstein (1905): light comes in packets, **photons**
- A blue photon carries more energy than a red one; more red photons do not help

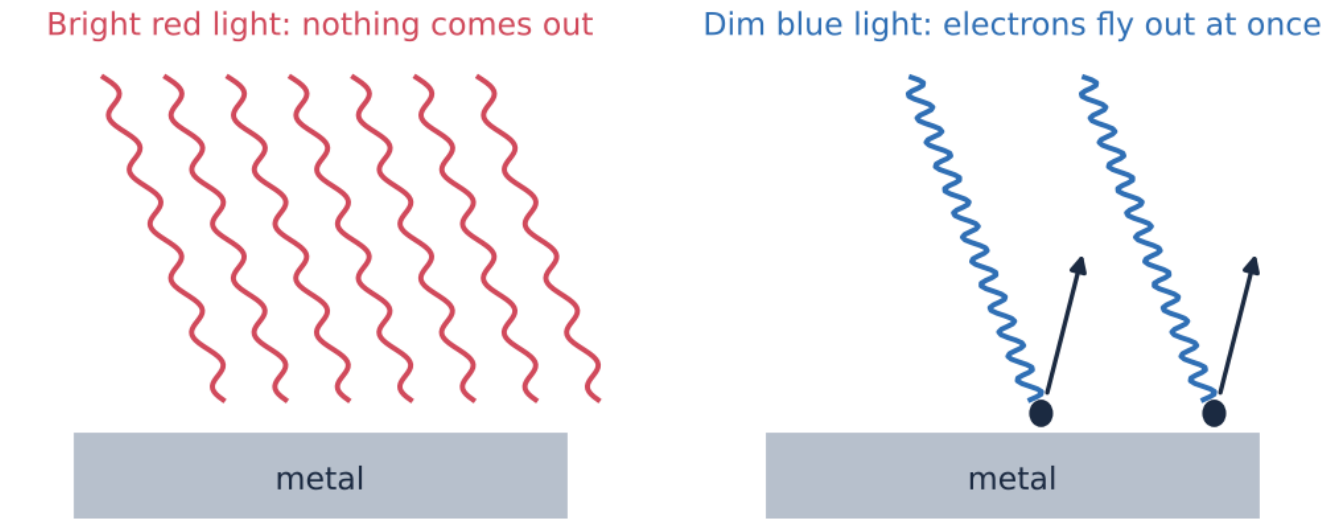


Figure made for this course.

Light also spreads like a wave

- Tiny balls through a gap would make **one spot**
- Light **spreads out** after a narrow gap, like ripples on a pond
- This is **diffraction**, a sure sign of a wave
- Light travels like a wave and arrives like a particle: **wave–particle duality**

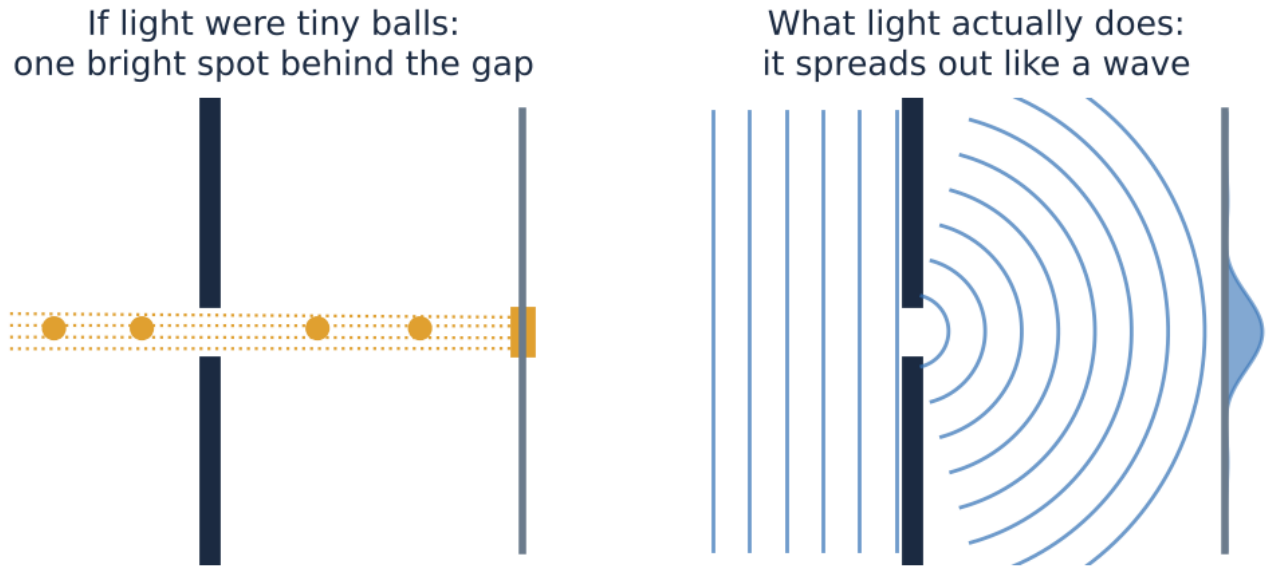


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Everything that moves has a wavelength

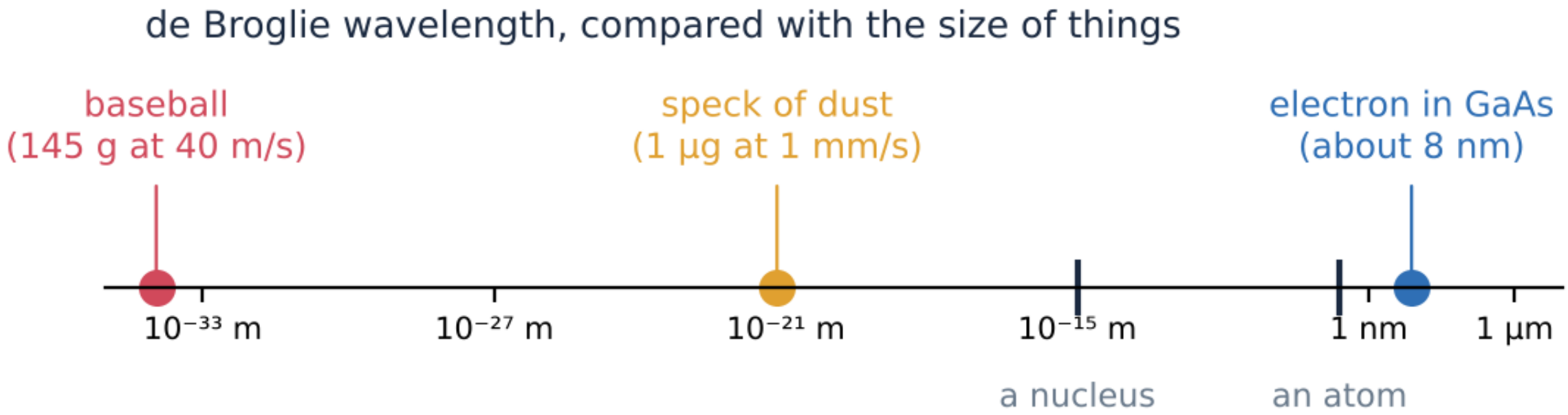
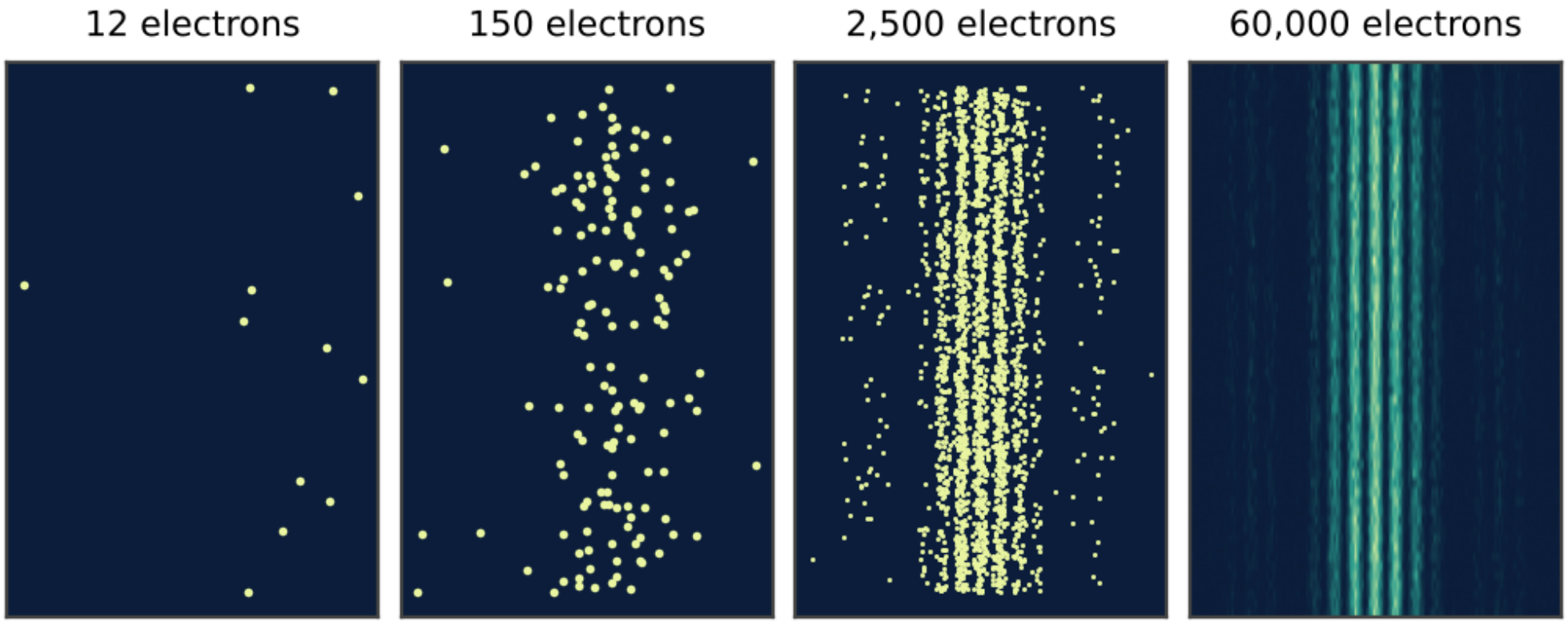


Figure made for this course.

De Broglie (1924): wavelength = h / momentum. Heavy, fast things have tiny wavelengths; an electron's is a few nanometres, the size of a transistor.

Electrons interfere one at a time



Simulated after Tonomura et al. (1989). Figure made for this course.

Each electron lands as a **dot** (a particle), but the dots build up the **stripes of a wave**. The wave sets the chances; the particle is what we detect.

3

The Schrödinger equation

It looks fearsome, but it is just energy bookkeeping.

Energy bookkeeping

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi$$

kinetic energy (how much the wave wiggles)

potential energy (the landscape)

total energy

Figure made for this course.

Kinetic energy + potential energy = total energy. A tightly wiggled wave has more kinetic energy. Only certain wave shapes balance the books: the **allowed energies**.

What the wave tells us

Where it is likely to be

We never see the wave ψ itself. The chance of finding the electron at a place follows ψ^2, the wave squared (Born).

A built-in fuzziness

A wave squeezed into a small space must mix many wavelengths, so position and speed cannot both be known exactly (Heisenberg).

4

An electron in a box

Only certain notes fit.

Only certain notes fit

- A guitar string is pinned at both ends, so only **whole numbers of humps** fit
- A trapped electron's wave must also **fit inside the box**
- So only **certain energies** are allowed: the energy is quantised
- More humps → tighter wiggles → **more energy**: 1, 4, 9, …

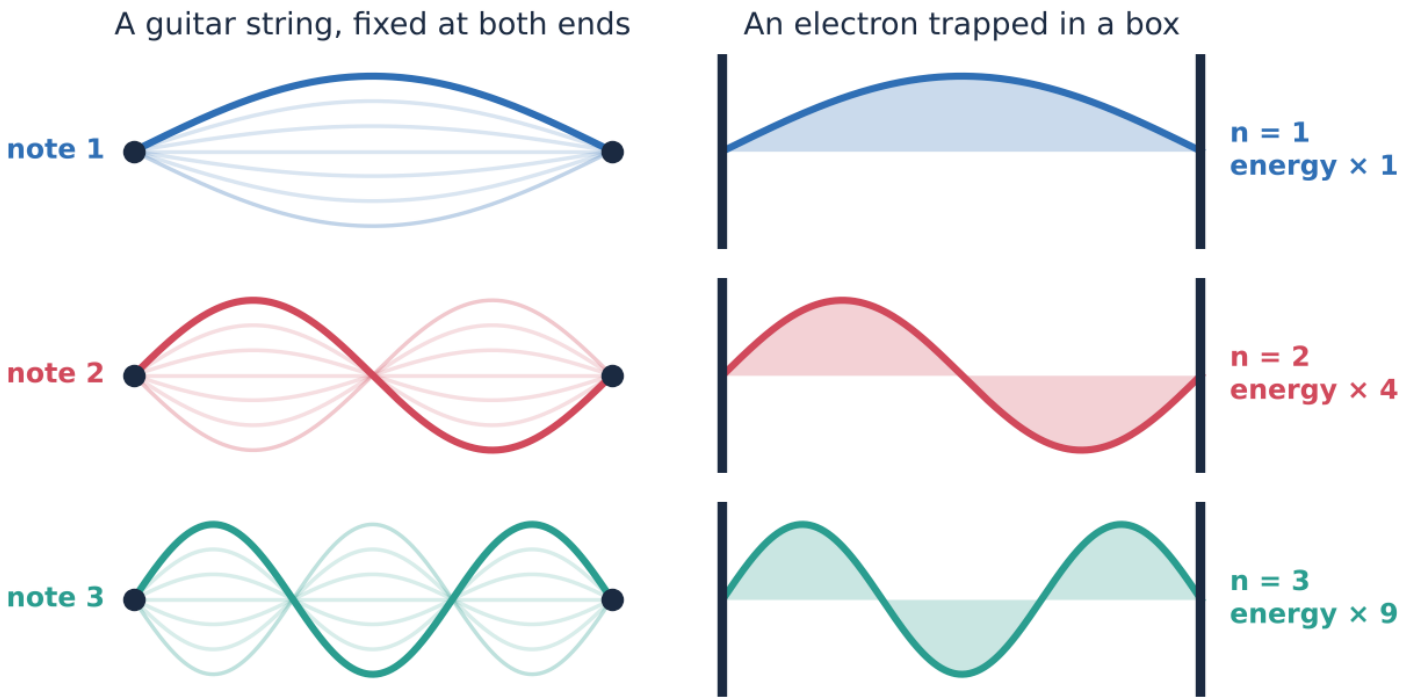


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What sets the energies

1 : 4 : 9

The allowed energies climb as the square of the number of humps

× 4

Halve the width of the box and every energy goes up four times

> 0

There is no n = 0: a trapped electron can never be completely still

Lighter particles have more widely spaced energies. Inside a semiconductor the electron acts lighter (about one-fifteenth in GaAs).

Where is the electron?

- The chance of finding it follows **ψ^2**
- Lowest state: most often **in the middle**
- Second state: **never** found exactly in the middle
- Each look finds the **whole electron** at one place; the curve is the pattern over many looks

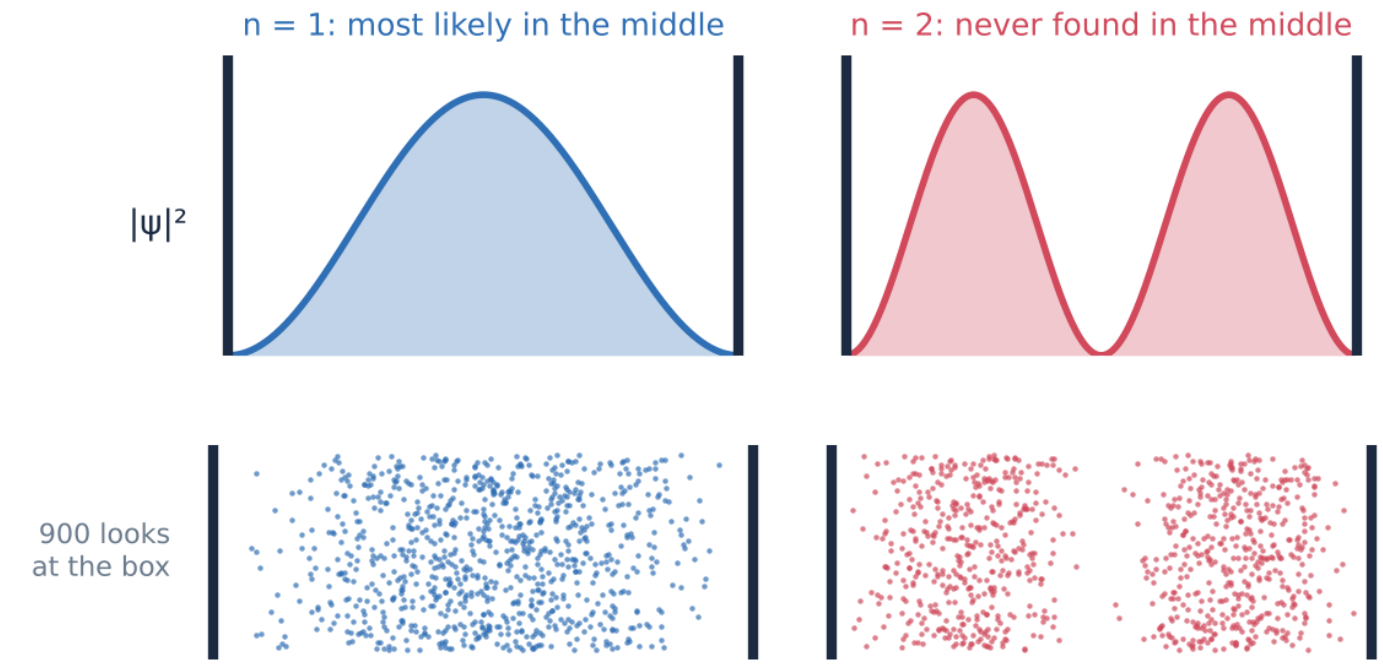


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Real walls leak a little

- Real walls: a thin layer of one semiconductor inside another
- The wave **leaks** a short way into the walls
- More room → **lower energies**: 32 meV instead of 56 for a 10 nm GaAs box
- A shallow box holds only a **few** levels
- The leaking wave is the start of **tunnelling** (Lecture 3)

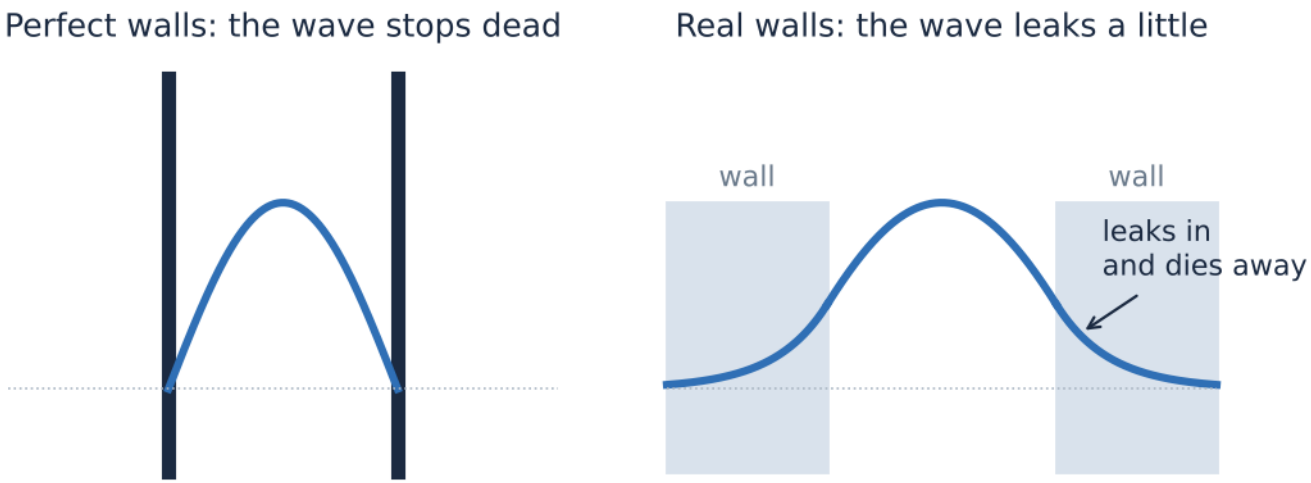


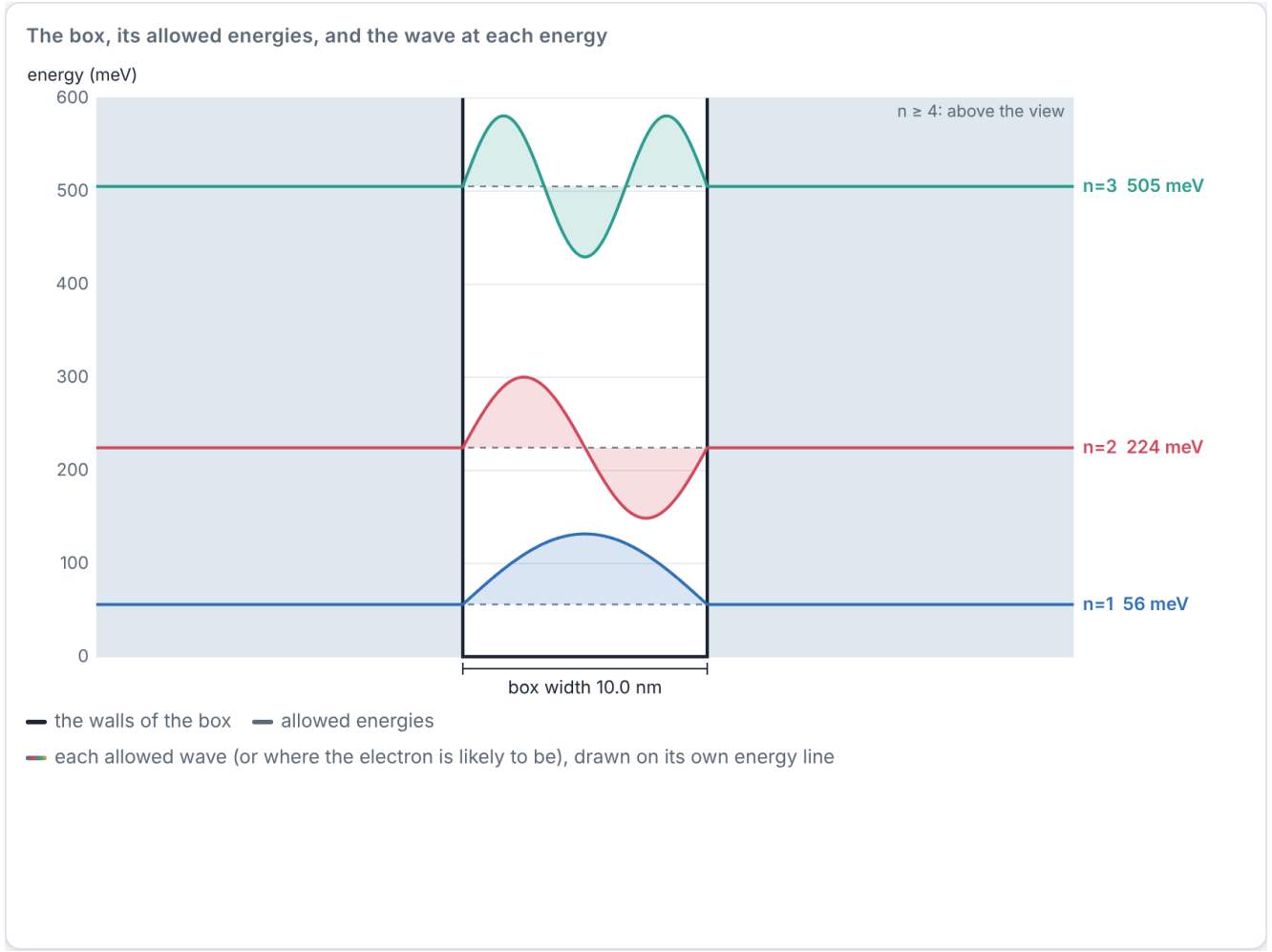
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Interactive animation 1

Particle in a box

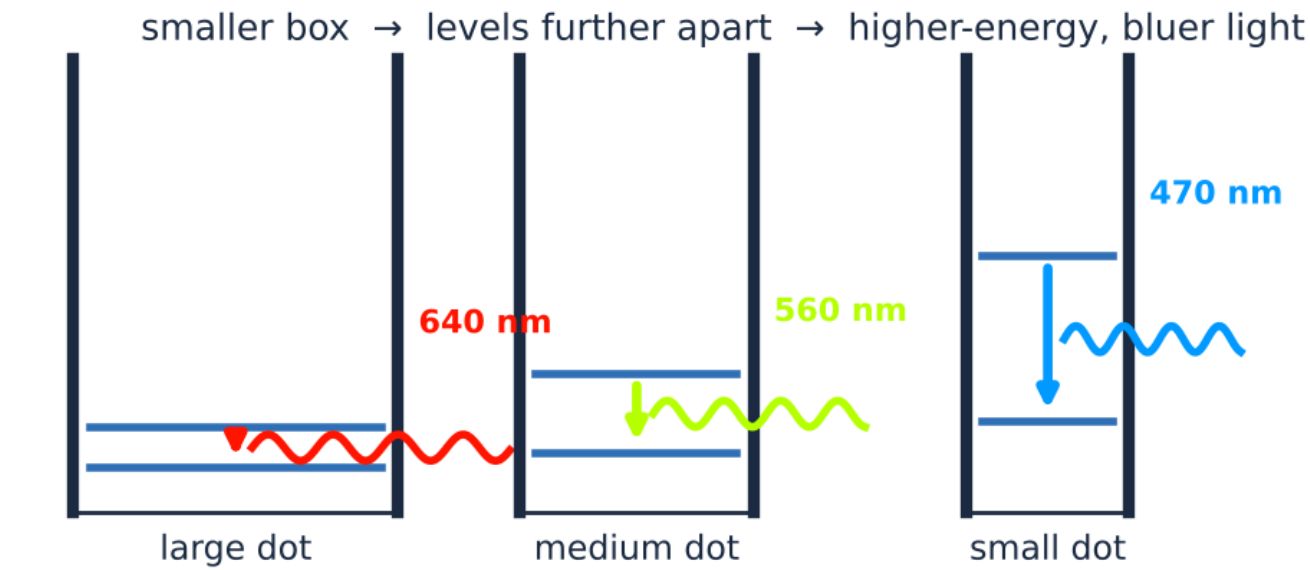
faizverse.github.io/nanoelectronic-devices/L2/particle-in-a-box.html

- Halve the box from 10 nm to 5 nm: predict, then watch
- Switch to 'where it is likely': where is n = 2 never found?
- Switch to leaky walls: do the energies go up or down?
- Watch the waves vibrate, higher ones faster



Smaller box, bluer light

- Squeeze the box and every level **moves up and apart**
- An electron dropping between levels gives out **light**
- Bigger gap → **higher-energy, bluer** light
- Same material, different size → **different colour**



Levels drawn schematically. Figure made for this course.

5

From a box to a real semiconductor

When do the levels actually show?

Inside a real crystal

A lighter electron

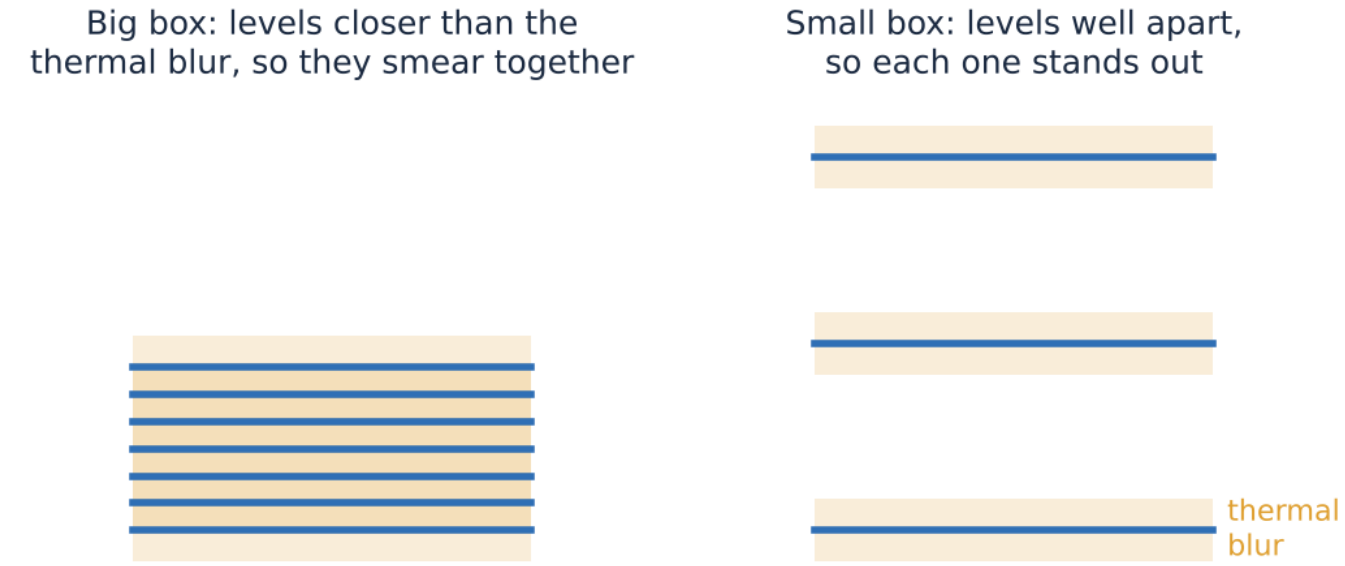
A perfect crystal lattice does not scatter the electron; its whole effect is to make it act lighter, the effective mass. About one-fifteenth in GaAs, a quarter in silicon.

Making the box

Sandwich a thin GaAs layer between AlGaAs layers. The outer layers are the walls; the layer's thickness is the width of the box. This is a quantum well.

When do the levels show?

- At room temperature every electron is jostled by about **26 meV**
- Levels closer than this blur **merge**
- Levels well apart **stand out**
- In GaAs: boxes narrower than about **15–20 nm**
- So the structures of Lecture 2 are a few to a few tens of nanometres



Drawn schematically. Figure made for this course.

Part 1 in six points

1 Waves and particles

Light and electrons are both.

2 Wavelength

Bigger momentum, shorter wavelength. Electrons: a few nm.

3 Schrödinger

Energy bookkeeping; ψ^2 gives where the electron is likely to be.

4 Trapped waves

Only certain energies, like a string's notes, never zero.

5 Size sets energy

Smaller box, further apart: why dot colour depends on size.

6 When it shows

When levels are further apart than the 26 meV thermal blur.

Credits

Made for this course

All figures except the photograph on slide 3 were made for KKKC6353 (scripts in figures-source/). The two-slit figure is simulated after Tonomura et al. (1989). Animation screenshot from faizverse.github.io/nanoelectronic-devices.

Sources

Quantum-dot photograph: Antipoff, Wikimedia Commons (CC BY-SA 3.0). Videos: NIBIB, 'What are quantum dots?'; Up and Atom, 'What is the Schrödinger equation, exactly?'. Further reading: Feynman, QED (1985); Griffiths and Schroeter (2018).